

CSE 309 Theory of Automata, Spring 2026
Exercise 7

1. **Decidability.** Prove whether each of the following language is Decidable, Turing-Recognizable, co-Turing Recognizable or None of them.
 - (a) $\{\langle M, r \rangle \mid M \text{ is a TM, } r \text{ is a regular expression, and } L(M) = L(r)\}$
 - (b) $\{\langle M_1, M_2, w \rangle \mid M_1 \text{ halts on } w \text{ if and only if TM } M_2 \text{ halts on } w\}$
 - (c) $\{\langle M, w \rangle \mid M \text{ halts on } w \text{ after at most } 2^{|w|} \text{ steps}\}$
 - (d) $\{\langle M \rangle \mid \text{there is a string } w \text{ on which TM } M \text{ halts after at most } 2^{|w|} \text{ steps}\}$
 - (e) $\{\langle M \rangle \mid \text{there is a non-empty language } L \text{ such that } L \cap L(M) = \emptyset\}$
 - (f) $\{w \mid w = \langle M \rangle \text{ for some TM } M\}$